Precision Tests of the Standard Model

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Standard Model Course Presentation

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Outline (Theory + Experiment)

1. Topics of the talk
2. Z physics
3. W mass
4. Precision global fit of electroweak parameters

Part 1: What we will (not) be covering

Primary Objectives:

- Define the foundational parameter space of the SM.
- Give current experimental values.
- What we will (not) be covering.

Standard Model Parameters

The Standard Model is defined by 19 free parameters (excluding neutrino masses), which must be determined experimentally.

Sector	**Parameter**	**Measured Value (Approx)**
Gauge Couplings	g_s, g, g'	$\alpha_s \approx 0.118, \alpha \approx 1/137, sin^2\theta_W \approx 0.223$
Higgs Sector	v, m_h	246 GeV, 125.25 GeV
Lepton Masses	m_e, m_μ, m_τ	0.511 MeV, 105.6 MeV, 1.77 GeV
Quark Masses	$m_u, m_d, \ldots, m_t$	$m_t \approx 172.7$ GeV
CKM Parameters	$\theta_{12}, \theta_{23}, \theta_{13}, \delta_{CP}$	(Measured Mixing Matrix)
Strong CP Phase	θ_{QCD}	$\lesssim 10^{-10}$

Goal:

1. *We will use three exquisitely measured inputs (G_F, M_Z (will see how to measure), and α) to predict and test the rest of the electroweak sector.*
2. *Measure the mass of the top quark directly and indirectly*

Part 2: Z physics

Primary Objectives:

- Derive the neutral current and Z decay rate.
- Extract the number of active neutrino generations (N_ν).
- Measure M_Z and the Forward-Backward Asymmetry (A_{FB}).

Neutral Current: Electroweak Mixing

1. Before Symmetry Breaking ($SU(2)_L \times U(1)_Y$):
The neutral gauge bosons W^3 and B couple to weak isospin (T_3) and hypercharge (Y):

$$\mathcal{L}_{NC} = \bar{\psi}\gamma^\mu \left(g T_3 W^3_\mu + g' \frac{Y}{2} B_\mu \right) \psi$$

2. Electroweak Rotation:
The fields mix via the Weinberg angle (θ_W):

$$W^3_\mu = \cos\theta_W Z_\mu + \sin\theta_W A_\mu \quad \text{and} \quad B_\mu = -\sin\theta_W Z_\mu + \cos\theta_W A_\mu$$

3. The Physical Z Boson Interaction:
Substituting the fields and using $Q = T_3 + Y/2$ isolates the Z coupling:

$$\mathcal{L}_Z = \frac{g}{\cos\theta_W} \bar{\psi}\gamma^\mu \left(T_3 - Q \sin^2\theta_W \right) \psi Z_\mu$$

Chiral Decomposition ($v_f - a_f\gamma^5$): $\quad v_f = T_3^f - 2Q_f \sin^2\theta_W \quad$ and $\quad a_f = T_3^f$

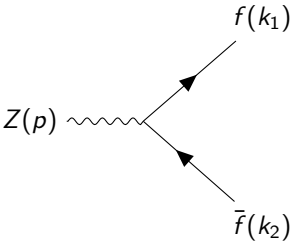
Deriving the Decay Rate: $Z \to f\bar{f}$

1. The Matrix Element ($\mathcal{M}$):

$$\mathcal{M} = -i\frac{g}{2\cos\theta_W}\epsilon_\mu(p)\bar{u}(k_1)\gamma^\mu(v_f - a_f\gamma^5)v(k_2)$$

2. Spin-Averaged Squared Amplitude:
(In the massless limit $m_f \to 0$)

$$|\overline{\mathcal{M}}|^2 = \frac{g^2 M_Z^2}{3\cos^2\theta_W}(v_f^2 + a_f^2)N_c$$



3. Final Partial Decay Width (Γ):

$$\Gamma(Z \to f\bar{f}) = \frac{G_F M_Z^3}{6\sqrt{2}\pi}(v_f^2 + a_f^2)N_c$$

Extraction of M_Z, Γ_Z, and N_ν

1. The Z^0 Mass and Width

Measured via an **Energy Scan**, the cross-section traces a precise **Breit-Wigner** resonance curve:

$$\sigma(s) \propto \frac{s}{(s - M_Z^2)^2 + M_Z^2 \Gamma_Z^2}$$

Fitting this curve extracts two independent parameters:

- M_Z **(Peak Position):** 91.1876 ± 0.0021 GeV.
- Γ_Z **(Peak Width):** 2.4952 ± 0.0023 GeV.

2. Number of Neutrinos (N_ν)

The total decay width (Γ_Z, measured on the left) is the sum of partial widths:

$$\Gamma_Z = \Gamma_{\text{had}} + 3\Gamma_l + N_\nu \Gamma_\nu$$

Solving for the invisible width gives:

$$N_\nu = \frac{\Gamma_Z^{\text{meas}} - \Gamma_{\text{had}}^{\text{meas}} - 3\Gamma_l^{\text{meas}}}{\Gamma_\nu^{\text{theo}}}$$

Theoretical Neutrino Width (Γ_ν):

$$\Gamma_\nu = \frac{G_F M_Z^3}{6\sqrt{2}\pi}\left[\left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2\right] = \frac{G_F M_Z^3}{12\sqrt{2}\pi} \approx 167\text{MeV}$$

- Result: 2.984 ± 0.008

Angular Distribution and A_{FB}

1. Differential Cross-Section:
The angular distribution of $e^+e^- \to \mu^+\mu^-$ is:

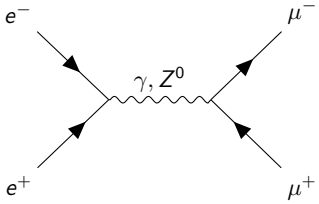
$$\frac{d\sigma}{d\cos\theta} \propto \underbrace{(v_e^2+a_e^2)^2(1+\cos^2\theta)}_{\text{Symmetric}} + \underbrace{8(v_e a_e)^2\cos\theta}_{\text{Asymmetric}}$$

2. The Asymmetry Observable:
Integrating over the Forward ($\cos\theta > 0$) and Backward ($\cos\theta < 0$) hemispheres:

$$A_{FB} = \frac{\sigma_F - \sigma_B}{\sigma_F + \sigma_B} = \frac{3}{4}A_e A_\mu$$

Where $A_f = \frac{2v_f a_f}{v_f^2+a_f^2}$. *This isolates* $\sin^2\theta_W$.

Interference: γ and Z^0



From A_{FB}, LEP measured:
$\sin^2\theta_W \approx 0.2315$

Which predicts a W mass:
$M_W = M_Z\cos\theta_W \approx$ **79.94** GeV

Part 3: W mass

Primary Objectives:

- Charged current Lagrangian showing (V-A) structure.
- Measuring the mass of W boson.

Gauge Origin of Charged Currents

1. The Unbroken $SU(2)_L$ Interaction:
Before symmetry breaking, the left-handed fermion doublets L couple to the triplet of massless $SU(2)$ gauge fields W^a_μ via the generators $T^a = \sigma^a/2$:

$$\mathcal{L}_{SU(2)} = \bar{L}\gamma^\mu \left(i\partial_\mu - g\frac{\sigma^a}{2} W^a_\mu \right) L$$

2. Defining the Physical $W^\pm$ Fields:
To identify the charge-carrying bosons, we define the ladder operators from the W^1 and W^2 Cartesian components:

$$W^\pm_\mu = \frac{1}{\sqrt{2}}(W^1_\mu \mp iW^2_\mu)$$

3. The Post-EWSB Lagrangian:
After the Higgs mechanism, the W bosons acquire mass $M_W = \frac{1}{2}gv$. The interaction is projected into the physical "V-A" charged current:

$$\mathcal{L}_{CC} = \frac{g}{\sqrt{2}} \left[\bar{u}_i \gamma^\mu \frac{(1-\gamma^5)}{2} V_{ij} d_j W^+_\mu + \text{h.c.} \right]$$

Measuring the W Mass: The LEP-II Era

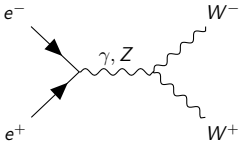
1. Crossing the Threshold:
LEP-I sat at $\sqrt{s} \approx 91$ GeV (the Z pole). To produce W bosons, charge conservation requires W^+W^- pair production.

LEP-II doubled the beam energy to cross the kinematic threshold: $\sqrt{s} \geq 2M_W \approx 161.2$ GeV.

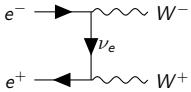
2. The Measurement Techniques:

- **Threshold Scan:** The $e^+e^- \to W^+W^-$ cross-section near 161 GeV is extremely sensitive to M_W.
- **Direct Kinematic Reconstruction:** At $\sqrt{s} = 172$–209 GeV, fully reconstructing the invariant mass of the decay products:
  - Fully Hadronic ($q\bar{q}q\bar{q}$)
  - Semi-Leptonic ($q\bar{q}\ell\nu$)
  - *Fully Leptonic is difficult due to missing energy.*

Dominant LEP-II Channels



s-channel (Gauge TGC)



t-channel (ν_e exchange)

The high-energy unitarity of this process is strictly enforced by a delicate cancellation between these channels.

Combined LEP-II Precision: 80.376 ± 0.033 GeV

Kinematics of W Pair Production and Decay

1. Center-of-Mass Frame (LEP-II):

The initial state is completely known:
$p_{e^-} + p_{e^+} = (\sqrt{s}, \vec{0})$.
By symmetry, each W carries exactly beam energy:
$E_W = \frac{\sqrt{s}}{2}$.

2. Invariant Mass Reconstruction:

Consider a W decaying into two massless fermions f_1 and $\bar{f}_2$ with 4-momenta p_1 and p_2.

Theoretical Mass Relation

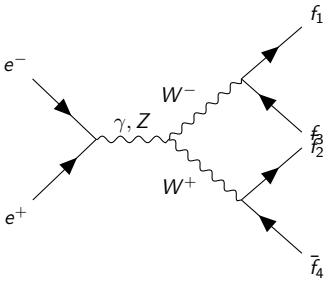
$$M_W^2 = (p_1 + p_2)^2 \approx 2E_1E_2(1 - \cos\theta_{12})$$

Where θ_{12} is the opening angle between the two decay products.

3. The Kinematic Constraints:

$$\sum_{i=1}^{4} E_i = \sqrt{s} \quad \text{and} \quad \sum_{i=1}^{4} \vec{p}_i = 0$$

Four-Fermion Final State



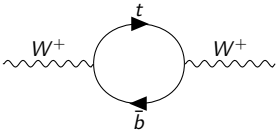
Part 4: Precision global fit of electroweak parameters

Primary Objectives:

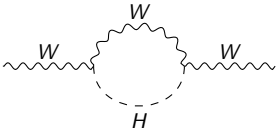
- Connecting G_F, α, and M_Z to M_W via quantum loops (Δr).
- Show how global fits successfully predicted(put bounds on) the Top and Higgs masses.
- Reconstructing the Top Quark at the Tevatron.

Radiative Corrections: Vacuum Polarization

Fermion Loops



Boson Loops



The Tree-Level Baseline:
At tree level, the mass ratio is locked at exactly 1:

$$\rho = \frac{M_W^2}{M_Z^2 \cos^2 \theta_W} = 1$$

The Top Quark Correction ($\Delta\rho$):
Quantum loops shift these masses. For the W boson, the dominant effect comes from the $t\bar{b}$ loop. Because the top is vastly heavier than the bottom ($m_t \gg m_b$), this mass splitting generates a massive **quadratic** correction to the ρ parameter:

$$\Delta\rho \approx \frac{3G_F}{8\sqrt{2}\pi^2} m_t^2$$

The Global Fit (Δr)

To calculate the true W mass, theorists combine all loop effects into a single total radiative correction parameter, Δr.

Relating the Parameters:

$$\Delta r \approx \Delta\alpha - \frac{\cos^2\theta_W}{\sin^2\theta_W}\boldsymbol{\Delta\rho} + \Delta r_{\text{Higgs}}$$

($\Delta\alpha$ is the photon vacuum polarization, $\Delta\rho$ carries the m_t^2 dependence, and Δr_{Higgs} carries a $\ln(m_H)$ dependence.)

The Full Electroweak Relation

$$M_W^2\left(1 - \frac{M_W^2}{M_Z^2}\right) = \frac{\pi\alpha}{\sqrt{2}G_F}(1 + \Delta r)$$

The Predictive Power of the Global Fit:

- **Top Quark:** Because it enters as m_t^2, precision LEP measurements predicted $m_t \approx 170$ GeV.
- **Higgs Boson:** Because it only enters logarithmically ($\ln m_H$), the mass was harder to pin down, but fits constrained $m_H \lesssim 160$ GeV.

Kinematics of Top Quark Reconstruction (Tevatron)

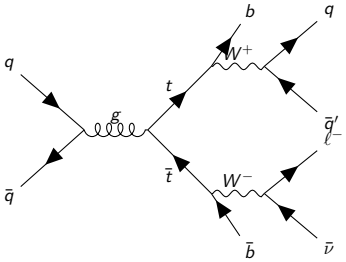
1. The Production and Decay:
$p\bar{p} \to t\bar{t}$ dominated by $q\bar{q}$ annihilation
$m_t > M_W + m_b$, so it decays before hadronizing:
$t \to Wb$.

2. The Lepton+Jets Topology:
The final = 4 jets $(q, \bar{q}', b, \bar{b})$, 1 lepton (ℓ), and missing transverse energy $(\vec{p}_T^{\,\nu})$

3. The Kinematic Constraints:

- **W-Mass Calibration:** $(p_q + p_{\bar{q}'})^2 = M_W^2$.
- **Neutrino p_z Recovery:** The lepton and missing energy must reconstruct the leptonic W:
  $(p_\ell + p_\nu)^2 = M_W^2 \implies$ *Yields quadratic for* p_z^ν.
- **Top Mass Equality:** Both branches must yield the same parent mass:
  $m_t^2 = (p_q + p_{\bar{q}'} + p_b)^2 = (p_\ell + p_\nu + p_{\bar{b}})^2$

The $t\bar{t}$ Golden Channel



Experimental Result:
Tevatron Combined: $m_t \approx 174.30 \pm 0.65$ GeV

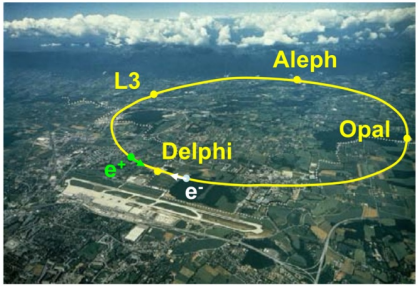
Current World Average: $m_t \approx 172.69 \pm 0.30$ GeV

Summary

- **Parameter Space:** The SM relies on experimental inputs (G_F, M_Z, α) to anchor its predictive framework.
- **Tree-Level Triumphs:** The Z pole measurements at LEP perfectly confirmed the $SU(2)_L \times U(1)_Y$ structure and definitively proved there are 3 neutrino generations.
- **Quantum Consistency:** The Standard Model is a fully renormalizable theory. Its loop structure (Δr) allowed physicists to map out the high-energy frontier and successfully predict the Top and Higgs masses purely from electroweak precision data.

LEP Collider

★The Large Electron Positron (LEP) Collider at CERN **(1989-2000)** was designed to make precise measurements of the properties of the Z and W bosons.



- 26 km circumference accelerator straddling French/Swiss boarder
- Electrons and positrons collided at 4 interaction points
- 4 large detector collaborations (each with 300-400 physicists):
  - **ALEPH**,
  - **DELPHI**,
  - **L3**,
  - **OPAL**

Basically a large Z and W factory:

- ★ **1989-1995:** Electron-Positron collisions at $\sqrt{s}$ = **91.2 GeV**
  - 17 Million Z bosons detected
- ★ **1996-2000:** Electron-Positron collisions at $\sqrt{s}$ = **161-208 GeV**
  - 30000 W^+W^- events detected

Cross Section Measurement

★ To work out cross sections, first count events of each type

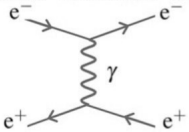
★ Then need to know "integrated luminosity" of colliding beams, i.e. the relation between cross-section and expected number of interactions

$$N_{\text{events}} = \mathscr{L}\sigma$$

★ To calculate the integrated luminosity need to know numbers of electrons and positrons in the colliding beams and the exact beam profile

- very difficult to achieve with precision of better than 10%

★ Instead "normalise" using another type of event:



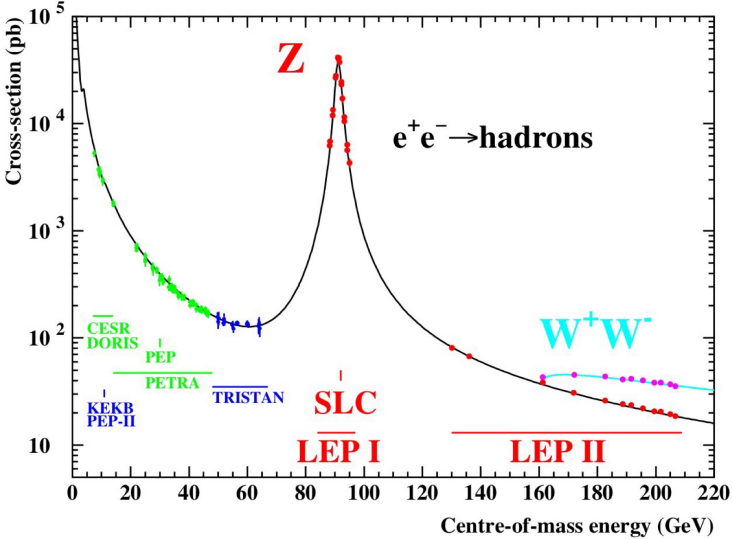
- Use the QED Bhabha scattering process
- QED, so cross section can be calculated very precisely
- Very large cross section – small statistical errors
- Reaction is very forward peaked – i.e. the electron tends not to get deflected much

- Count events where the electron is scattered in the very forward direction

$$N_{\text{Bhabha}} = \mathscr{L}\sigma_{\text{Bhabha}}$$

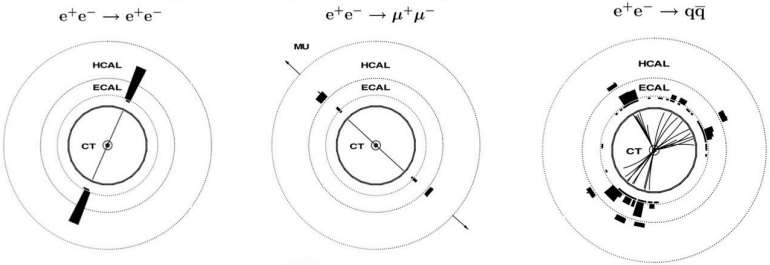
★ Hence all other cross sections can be expressed as $\boxed{\sigma_i = \frac{N_i}{N_{\text{Bhabha}}}\sigma_{\text{Bhabha}}}$

e^-e^+ Annihilation

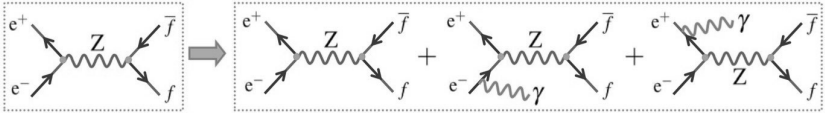


Measurement of Z Mass

★ Each has a distinct topology in the detectors, e.g.



★ In practise, it is not that simple, QED corrections distort the measured line-shape
★ One particularly important correction: initial state radiation (**ISR**)



Measurement of Z Mass

★ Initial state radiation reduces the centre-of-mass energy of the e^+e^- collision

$$e^+ \xrightarrow{E} \quad \xleftarrow{E} e^- \qquad \sqrt{s} = 2E$$

becomes

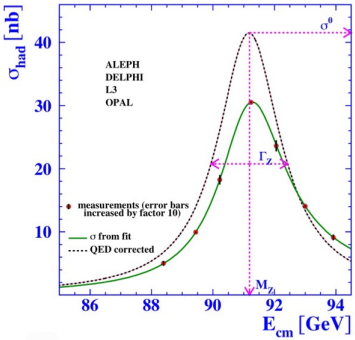
$$\xrightarrow{E} \quad \xleftarrow{E - E_\gamma} \qquad \sqrt{s'} \approx 2E\left(1 - \frac{E_\gamma}{2E}\right)$$

★ Measured cross section can be written:

$$\sigma_{\text{meas}}(E) = \int \sigma(E') f(E', E) \mathrm{d}E'$$

Probability of e+e- colliding with C.o.M. energy E' when C.o.M energy before radiation is E

★ Fortunately can calculate $f(E', E)$ very precisely, just QED, and can then obtain Z line-shape from measured cross section



★ In principle the measurement of m_Z and Γ_Z is rather simple: run accelerator at different energies, measure cross sections, account for ISR, then find peak and FWHM

$$m_Z = 91.1875 \pm 0.0021\,\text{GeV} \qquad \Gamma_Z = 2.4952 \pm 0.0023\,\text{GeV}$$

Number of Generations

- ★Total decay width measured from Z line-shape: $\Gamma_Z = 2.4952 \pm 0.0023\,\mathrm{GeV}$
- ★ If there were an additional 4th generation would expect $Z \to \nu_4 \bar{\nu}_4$ decays even if the charged leptons and fermions were too heavy (i.e. > $m_Z/2$)
- ★ Total decay width is the sum of the partial widths:

$$\Gamma_Z = \Gamma_{ee} + \Gamma_{\mu\mu} + \Gamma_{\tau\tau} + \Gamma_{\text{hadrons}} + \Gamma_{\nu_1\nu_1} + \Gamma_{\nu_2\nu_2} + \Gamma_{\nu_3\nu_3} + ?$$

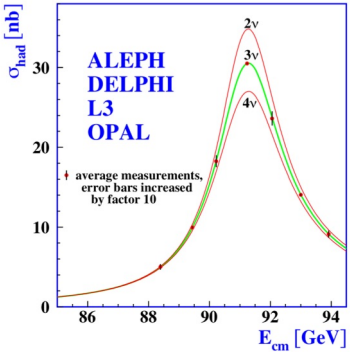
- ★ Although don't observe neutrinos, $Z \to \nu\bar{\nu}$ decays affect the Z resonance shape for all final states
- ★ For all other final states can determine partial decay widths from peak cross sections:

$$\sigma^0_{f\bar{f}} = \frac{12\pi}{m_Z^2}\frac{\Gamma_{ee}\Gamma_{ff}}{\Gamma_Z^2}$$

- ★ Assuming lepton universality:

$$\Gamma_Z = 3\Gamma_{\ell\ell} + \Gamma_{\text{hadrons}} + \boxed{N_\nu}\Gamma_{\nu\nu}$$

$$\boxed{N_\nu = 2.9840 \pm 0.0082}$$

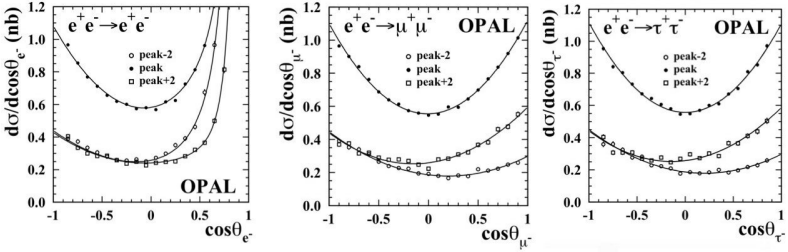


Measurement of Weinberg Angle

★ **The differential cross sections is therefore of the form:**

$$\frac{d\sigma}{d\Omega} = \kappa \times [A(1+\cos^2\theta) + B\cos\theta] \quad \begin{cases} A = [(c_L^e)^2 + (c_R^e)^2][(c_L^\mu)^2 + (c_R^\mu)^2] \\ B = [(c_L^e)^2 - (c_R^e)^2][(c_L^\mu)^2 - (c_R^\mu)^2] \end{cases}$$

$$A_{FB} = \frac{\sigma_F - \sigma_B}{\sigma_F + \sigma_B} = \frac{B}{(8/3)A} = \frac{3}{4}\left[\frac{(c_L^e)^2 - (c_R^e)^2}{(c_L^e)^2 + (c_R^e)^2}\right] \cdot \left[\frac{(c_L^\mu)^2 - (c_R^\mu)^2}{(c_L^\mu)^2 + (c_R^\mu)^2}\right]$$



$$A_{FB}^{e} = 0.0145 \pm 0.0025, \quad A_{FB}^{\mu} = 0.0169 \pm 0.0013 \quad \text{and} \quad A_{FB}^{\tau} = 0.0188 \pm 0.0017.$$

Measurement of Weinberg Angle

★ To relate these measurements to the couplings uses $A_{\rm FB} = \frac{3}{4} A_e A_\mu$

Putting everything together ⇨

$$\boxed{\begin{aligned} A_e &= 0.1514 \pm 0.0019 \\ A_\mu &= 0.1456 \pm 0.0091 \\ A_\tau &= 0.1449 \pm 0.0040 \end{aligned}}$$

with $$A_f \equiv \frac{2 c_V^f c_A^f}{(c_V^f)^2 + (c_A^f)^2} = 2 \frac{c_V/c_A}{1 + (c_V/c_A)^2}$$

★ Measured asymmetries give ratio of vector to axial-vector Z coupings.

★ In SM these are related to the weak mixing angle

$$\frac{c_V}{c_A} = \frac{I_W^3 - 2Q \sin^2\theta_W}{I_W^3} = 1 - \frac{2Q}{I_3} \sin^2\theta_W = 1 - 4|Q| \sin^2\theta_W$$

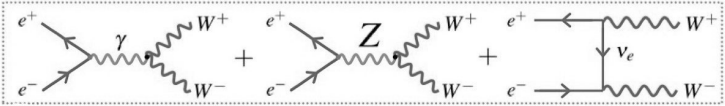
★ Asymmetry measurements give precise determination of $\sin^2\theta_W$

$$\boxed{\sin^2\theta_W = 0.23154 \pm 0.00016}$$

Measurement of W Mass

★ From 1995-2000 LEP operated above the threshold for W-pair production

★ Three diagrams

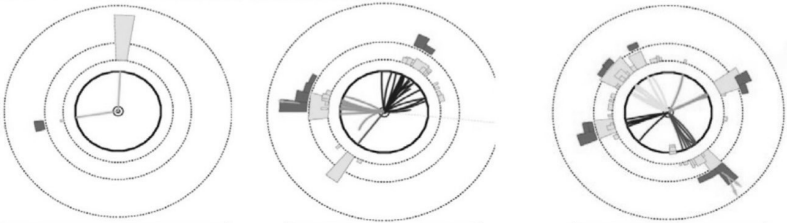


★ W bosons decay (p.459) either to leptons or hadrons with branching fractions:

$$Br(W^- \to \text{hadrons}) \approx 0.67 \qquad Br(W^- \to e^- \overline{\nu}_e) \approx 0.11$$

$$Br(W^- \to \mu^- \overline{\nu}_\mu) \approx 0.11 \qquad Br(W^- \to \tau^- \overline{\nu}_\tau) \approx 0.11$$

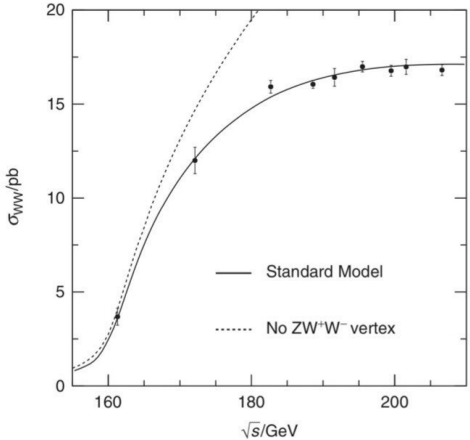
★ Gives rise to three distinct topologies



$W^+W^- \to \ell^+ \nu \ell^- \overline{\nu}$ | $W^+W^- \to q\overline{q}\ell\nu$ | $W^+W^- \to q\overline{q}q\overline{q}$

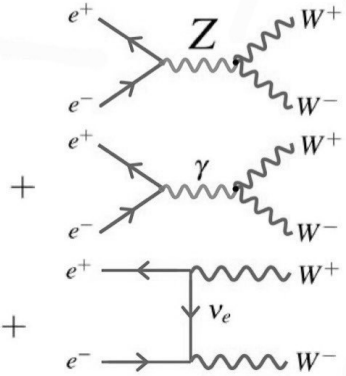
Measurement of W Mass

★ **Measure cross sections by counting events**



★ **Data consistent with SM expectation**

★ **Provides a direct test of ZW^+W^- vertex**



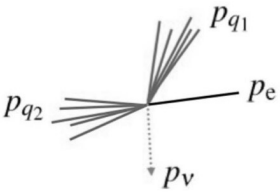
Measurement of W Mass

★ Unlike $e^+e^- \to Z$, the process $e^+e^- \to W^+W^-$ is not a resonant process

$\Longrightarrow$ Different method to measure W-boson Mass

•Measure energy and momenta of particles produced in the W boson decays, e.g.

$$W^+W^- \to q\bar{q}e^-\bar{\nu}$$



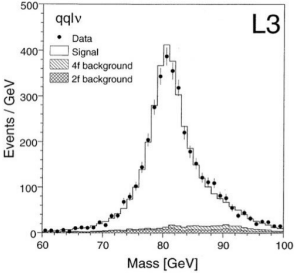
- Neutrino four-momentum from energy-momentum conservation !

$$p_{q_1} + p_{q_2} + p_e + p_\nu = (\sqrt{s}, 0)$$

- Reconstruct masses of two W bosons

$$M_+^2 = E^2 - \vec{p}^2 = (p_{q_1} + p_{q_2})^2$$
$$M_-^2 = E^2 - \vec{p}^2 = (p_e + p_\nu)^2$$

Measurement of W Mass



★ Peak of reconstructed mass distribution gives $m_W = 80.376 \pm 0.033\,\mathrm{GeV}$

★ Width of reconstructed mass distribution gives: $\Gamma_W = 2.196 \pm 0.083\,\mathrm{GeV}$

★ From LEP and elsewhere have precise measurements – can test predictions of the Standard Model !

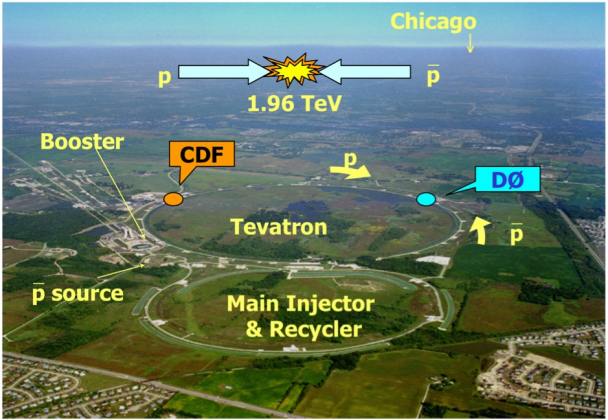
•e.g. predict: $m_W = m_Z \cos\theta_W$ measure $m_Z = 91.1875 \pm 0.0021\,\mathrm{GeV}$, $\sin^2\theta_W = 0.23154 \pm 0.00016$

•Therefore expect: $m_W = 79.946 \pm 0.008\,\mathrm{GeV}$ but measure $m_W = 80.376 \pm 0.033\,\mathrm{GeV}$

Tevatron Collider

★ The Tevatron collider at Fermilab (1992 - 2011) was mainly designed to discover top quark.

- 6.28 km synchrotron collider
- Collided proton and anti-proton at two interaction points
- Two detector collaborations:
  - CDF
  - D0
- 1992-1996: $\sqrt{s}$ = 1.8 TeV
- 2001-2011: $\sqrt{s}$ = 1.96 TeV

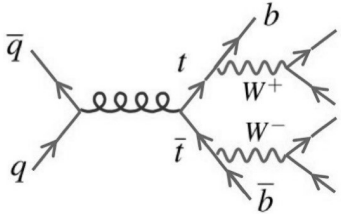


Measurement of Top Quark Mass

★ From virtual loop corrections and precise LEP data can predict the top quark mass:

$$\boxed{m_t^{\text{loop}} = 173 \pm 11\,\text{GeV}}$$

★ In 1994 top quark observed at the Tevatron proton anti-proton collider at Fermilab – with the predicted mass !



★ The top quark almost exclusively decays to a bottom quark since

$$|V_{tb}|^2 \gg |V_{td}|^2 + |V_{ts}|^2$$

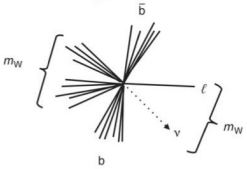
★ Complicated final state topologies:

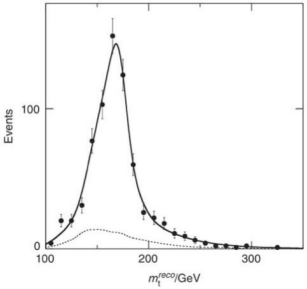
$$t\bar{t} \to b\bar{b}q\bar{q}q\bar{q} \to 6 \text{ jets}$$

$$t\bar{t} \to b\bar{b}q\bar{q}\ell\nu \to 4 \text{ jets} + \ell + \nu$$

$$t\bar{t} \to b\bar{b}\ell\nu\ell\nu \to 2 \text{ jets} + 2\ell + 2\nu$$

Measurement of Top Quark Mass





- b-quark jets identified using secondary vertices tagging
- Neutrino momentum is reconstructed using kinematic fitting.

$$m_t^{\text{meas}} = 174.2 \pm 3.3\,\text{GeV}$$

$$\Gamma_t = 2.0 \pm 0.6\,\text{GeV}$$

Thank You

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